SECTION 1. PRODUCT AND COMPANY IDENTIFICATION

Product name : POTENZOR®

Manufacturer or supplier's details

Company : FMC QUÍMICA DO BRASIL LTDA.

Address : AVENIDA DR. JOSÉ BONIFÁCIO COUTINHO NOGUEIRA 150 - 1º ANDAR - JARDIM MADALENA, CAMPINAS SP BRASIL
TELEFONE: (19) 2042.4500

Emergency telephone : Brazil: 0800 34 35 450 (24 hours)
+55-2139581449 (CHEMTREC)

Medical Emergency Number : 0800 7010 450

Recommended use of the chemical and restrictions on use

Recommended use : Herbicide

Restrictions on use : Use as recommended by the label.

SECTION 2. HAZARDS IDENTIFICATION

GHS Classification in accordance with ABNT NBR 14725 Standard

Skin sensitization : Category 1

Long-term (chronic) aquatic hazard : Category 2

GHS label elements in accordance with ABNT NBR 14725 Standard

Hazard pictograms :

Signal Word : WARNING

Hazard Statements : H317 May cause an allergic skin reaction.
H411 Toxic to aquatic life with long lasting effects.

Precautionary Statements :

Prevention:

P261 Avoid breathing mist or vapors.
P272 Contaminated work clothing should not be allowed out of the workplace.
P273 Avoid release to the environment.

P280 Wear protective gloves.

Response:

P302 + P352 IF ON SKIN: Wash with plenty of water.
P333 + P313 If skin irritation or rash occurs: Get medical advice/ attention.
P362 + P364 Take off contaminated clothing and wash it before reuse.
P391 Collect spillage.

Disposal:

P501 Dispose of contents/ container to an approved waste disposal plant.

Other hazards which do not result in classification

None known.

SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

Substance / Mixture : Mixture

Components

Chemical name	CAS-No.	Classification	Concentration (% w/w)
clomazone (ISO)	81777-89-1	Acute Tox. (Oral), 4 Acute Tox. (Dermal), 5 Aquatic Acute, 1 Aquatic Chronic, 1	>= 20 -< 25
sodium nitrate	7631-99-4	Ox. Sol., 2 Acute Tox. (Oral), 5 Serious eye damage/eye irritation, 2A Aquatic Acute, 3	>= 2,5 -< 5
Calcium chloride, dihydrate	10035-04-8	Acute Tox. (Oral), 5 Serious eye damage/eye irritation, 2A	>= 1 -< 5
Imazethapyr	81335-77-5	Acute Tox. (Oral), 5 Acute Tox. (Inhalation), 4 Acute Tox. (Dermal), 5 Aquatic Acute, 3 Aquatic Chronic, 3	>= 2,5 -< 5

SECTION 4. FIRST AID MEASURES

General advice : Move out of dangerous area.
Show this material safety data sheet to the doctor in attendance.
Do not leave the victim unattended.

If inhaled	:	If unconscious, place in recovery position and seek medical advice. If symptoms persist, call a physician.
In case of skin contact	:	Wash off with soap and water. If symptoms persist, call a physician. Wash contaminated clothing before re-use.
In case of eye contact	:	Flush eyes with water as a precaution. Remove contact lenses. Protect unharmed eye. Keep eye wide open while rinsing. If eye irritation persists, consult a specialist.
If swallowed	:	Keep respiratory tract clear. Do not give milk or alcoholic beverages. Never give anything by mouth to an unconscious person. If symptoms persist, call a physician.
Most important symptoms and effects, both acute and delayed	:	May cause an allergic skin reaction. Exposure to skin may result in mild symptoms include itching, hives or rash, and skin redness. More severe symptoms include sneezing, itchy watery eyes, and difficulty breathing.
Protection of first-aiders	:	Avoid inhalation, ingestion and contact with skin and eyes.
Notes to physician	:	Treat symptomatically.

SECTION 5. FIRE-FIGHTING MEASURES

Suitable extinguishing media	:	Dry chemical, CO_2, water spray or regular foam.
Unsuitable extinguishing media	:	Do not spread spilled material with high-pressure water streams.
Specific hazards during fire fighting	:	Do not allow run-off from fire fighting to enter drains or water courses.
Hazardous combustion products	:	Fire may produce irritating, corrosive and/or toxic gases. Nitrogen oxides (NOx) Sodium oxides Hydrogen cyanide Ammonia Carbon oxides
Specific extinguishing methods	:	Remove undamaged containers from fire area if it is safe to do so. Use a water spray to cool fully closed containers. Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. Collect contaminated fire extinguishing water separately. This must not be discharged into drains. Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations.

Special protective equipment for fire-fighters	:	Firefighters should wear protective clothing and self-contained breathing apparatus.

SECTION 6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures	:	Evacuate personnel to safe areas. Use personal protective equipment. If it can be safely done, stop the leak. Do not touch or walk through the spilled material. Avoid formation of aerosol.
Environmental precautions	:	Prevent product from entering drains. Prevent further leakage or spillage if safe to do so. If the product contaminates rivers and lakes or drains inform respective authorities.
Methods and materials for containment and cleaning up	:	Never return spills in original containers for re-use. Collect as much of the spill as possible with a suitable absorbent material. Pick up and transfer to properly labeled containers. Keep in suitable, closed containers for disposal.

SECTION 7. HANDLING AND STORAGE

Advice on protection against fire and explosion	:	Normal measures for preventive fire protection.
Advice on safe handling	:	Do not breathe vapors/dust. Avoid exposure - obtain special instructions before use. Avoid contact with skin and eyes. For personal protection see section 8. Smoking, eating and drinking should be prohibited in the application area. Dispose of rinse water in accordance with local and national regulations. Persons susceptible to skin sensitization problems or asthma, allergies, chronic or recurrent respiratory disease should not be employed in any process in which this mixture is being used.
Hygiene measures	:	Avoid contact with skin, eyes and clothing. Do not inhale aerosol. When using do not eat or drink. When using do not smoke. Wash hands before breaks and at the end of workday.
Conditions for safe storage	:	Keep container tightly closed in a dry and well-ventilated place. Containers which are opened must be carefully resealed and kept upright to prevent leakage. Electrical installations / working materials must comply with

		the technological safety standards.
Further information on storage stability	:	No decomposition if stored and applied as directed.

SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Ingredients with workplace control parameters

Contains no substances with occupational exposure limit values.

Personal protective equipment

Respiratory protection	:	In case of mist, spray or aerosol exposure wear suitable personal respiratory protection and protective suit.
Hand protection		
Material	:	Protective gloves
Remarks	:	The suitability for a specific workplace should be discussed with the producers of the protective gloves.
Eye protection	:	Eye wash bottle with pure water Tightly fitting safety goggles
Skin and body protection	:	Impervious clothing Choose body protection according to the amount and concentration of the dangerous substance at the work place.
Protective measures	:	Plan first aid action before beginning work with this product.

SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

Physical state	:	liquid
Color	:	green
Odor	:	characteristic
Odor Threshold	:	No data available
pH	:	6,68 (20 °C) (undiluted)
Melting point/ range	:	No data available
Boiling point/boiling range	:	85,6 °C
Flash point	:	No flash up to boiling point.

Evaporation rate	:	No data available
Self-ignition	:	No data available
Upper explosion limit / Upper flammability limit	:	No data available
Lower explosion limit / Lower flammability limit	:	No data available
Vapor pressure	:	No data available
Relative vapor density	:	No data available
Relative density	:	No data available
Density	:	1,1234 g/cm3 (20 °C) Method: OECD Test Guideline 109
Solubility(ies)		
Water solubility	:	No data available
Partition coefficient: n-octanol/water	:	No data available
Autoignition temperature	:	No data available
Decomposition temperature	:	No data available
Viscosity		
Viscosity, dynamic	:	122,8 mPa.s (20 °C) Method: OECD Test Guideline 114 92,85 mPa.s (40 °C) Method: OECD Test Guideline 114
Viscosity, kinematic	:	No data available
Explosive properties	:	Not explosive
Oxidizing properties	:	Non-oxidizing
Surface tension	:	74,53 mN/m, 1 g/l, 20 °C, OECD Test Guideline 115
Molecular weight	:	Not applicable
Metal corrosion rate	:	Not corrosive to metals.

SECTION 10. STABILITY AND REACTIVITY

Reactivity	:	No decomposition if stored and applied as directed.
Chemical stability	:	No decomposition if stored and applied as directed.
Possibility of hazardous reactions	:	No decomposition if stored and applied as directed.
Conditions to avoid	:	Avoid extreme temperatures. Avoid formation of aerosol.
Incompatible materials	:	Avoid strong acids, bases, and oxidizers.
Hazardous decomposition products	:	No hazardous decomposition products are known.

SECTION 11. TOXICOLOGICAL INFORMATION

Acute toxicity

Based on available data, the classification criteria are not met.

Product:

Acute oral toxicity	:	LD50 (Rat, female): > 2.000 mg/kg Method: OECD Test Guideline 423 GLP: yes Assessment: The substance or mixture has no acute oral toxicity Remarks: no mortality
Acute inhalation toxicity	:	LC50 (Rat): > 3,056 mg/l Exposure time: 4 h Test atmosphere: dust/mist Method: OECD Test Guideline 403 GLP: yes Assessment: The substance or mixture has no acute inhalation toxicity Remarks: no mortality
Acute dermal toxicity	:	(Rat, female): > 2.000 mg/kg Method: OECD Test Guideline 402 GLP: yes Remarks: no mortality Acute toxicity estimate: > 5.000 mg/kg Method: Calculation method

Components:

clomazone (ISO):

Acute oral toxicity	:	LD50 (Rat, female): 768 mg/kg Method: OECD Test Guideline 425 LD50 (Rat, female): 300 - 2.000 mg/kg Method: OECD Test Guideline 423

		Target Organs: Liver Assessment: The component/mixture is moderately toxic after single ingestion.
		LD50 (Rat, female): 1.564 mg/kg Symptoms: ataxia
Acute inhalation toxicity	:	LC50 (Rat, male and female): > 12,1 mg/l Exposure time: 4 h Test atmosphere: dust/mist Method: OECD Test Guideline 403 Symptoms: apathy Assessment: The substance or mixture has no acute inhalation toxicity Remarks: no mortality
		LC50 (Rat): > 7,4 mg/l Exposure time: 4 h Test atmosphere: dust/mist Method: OECD Test Guideline 403 Assessment: The substance or mixture has no acute inhalation toxicity
Acute dermal toxicity	:	LD50 (Rabbit, male and female): > 2.000 mg/kg Method: US EPA Test Guideline OPP 81-2 Assessment: The component/mixture is minimally toxic after single contact with skin. Remarks: no mortality
		LD50 (Rabbit, male and female): > 4.000 mg/kg Method: OECD Test Guideline 402 Remarks: no mortality

sodium nitrate:

Acute oral toxicity	:	LD50 (Rat, male and female): 3.430 mg/kg Method: OECD Test Guideline 401
		LD50 (Rat): > 2.000 mg/kg Method: OECD Test Guideline 425
Acute inhalation toxicity	:	LD50 (Rat): > 0,527 mg/l Exposure time: 4 h Test atmosphere: dust/mist
Acute dermal toxicity	:	LD50 (Rat, male and female): > 5.000 mg/kg Method: OECD Test Guideline 402

Calcium chloride, dihydrate:

Acute oral toxicity	:	LD50 (Rat, male): 2.120 mg/kg Method: OECD Test Guideline 401 Remarks: mortality
		LD50 (Rat, female): 2.361 mg/kg Method: OECD Test Guideline 401

Remarks: mortality

LD50 (Rat, male and female): 2.301 mg/kg
Method: OECD Test Guideline 401
Symptoms: Lethargy, Necrosis, Gastrointestinal disturbance, respiratory tract irritation
Remarks: mortality

Acute dermal toxicity : LD50 (Rabbit, male and female): > 5.000 mg/kg
Remarks: no mortality

Imazethapyr:

Acute oral toxicity : LD50 (Rat): > 2.000 mg/kg
Method: OECD Test Guideline 423
Assessment: The component/mixture is minimally toxic after single ingestion.

Acute inhalation toxicity : LC50 (Rat): > 4,3 mg/l
Exposure time: 4 h
Test atmosphere: dust/mist
Method: OECD Test Guideline 403

Acute dermal toxicity : LD50 (Rat): > 2.000 mg/kg
Method: OECD Test Guideline 402
Assessment: The component/mixture is minimally toxic after single contact with skin.

Skin corrosion/irritation

Based on available data, the classification criteria are not met.

Product:

Species : reconstructed human epidermis (RhE)
Assessment : Not classified as irritant
Method : OECD Test Guideline 439
GLP : yes

Components:

clomazone (ISO):

Species : Rabbit
Assessment : Not classified as irritant
Method : OECD Test Guideline 404
Result : slight or no skin irritation.

Calcium chloride, dihydrate:

Species : Rabbit
Method : OECD Test Guideline 404
Result : No skin irritation

Imazethapyr:

Species : Rabbit
Assessment : Not classified as irritant

Method	:	OECD Test Guideline 404
Result	:	slight irritation

Serious eye damage/eye irritation

Based on available data, the classification criteria are not met.

Product:

Species	:	Rabbit
Assessment	:	No eye irritation
Method	:	OECD Test Guideline 405
GLP	:	yes
Species	:	Bovine cornea
Method	:	OECD Test Guideline 437
Remarks	:	not corrosive
Remarks	:	Vapors may cause irritation to the eyes, respiratory system and the skin.

Components:

clomazone (ISO):

Species	:	Rabbit
Result	:	Slight or no eye irritation
Assessment	:	Not classified as irritant
Method	:	OECD Test Guideline 405
GLP	:	yes

sodium nitrate:

Species	:	Rabbit
Result	:	Eye irritation
Assessment	:	Irritating to eyes.
Method	:	OECD Test Guideline 405

Calcium chloride, dihydrate:

Species	:	Rabbit
Result	:	Irritation to eyes, reversing within 21 days
Method	:	OECD Test Guideline 405

Imazethapyr:

Species	:	Rabbit
Result	:	slight irritation
Assessment	:	Not classified as irritant
Method	:	OECD Test Guideline 405

Respiratory or skin sensitization

Skin sensitization

May cause an allergic skin reaction.

Respiratory sensitization

Based on available data, the classification criteria are not met.

Product:

Test Type	:	Local lymph node test
Routes of exposure	:	Dermal
Assessment	:	May cause sensitization by skin contact.
Method	:	OECD Test Guideline 442B
GLP	:	yes
Remarks	:	Causes sensitization.

Components:

clomazone (ISO):

Test Type	:	Buehler Test
Species	:	Guinea pig
Assessment	:	Not a skin sensitizer.
Method	:	OECD Test Guideline 406
Result	:	Not a skin sensitizer.
GLP	:	yes
Species	:	Guinea pig
Assessment	:	Not a skin sensitizer.
Method	:	US EPA Test Guideline OPP 81-6
Result	:	Not a skin sensitizer.

sodium nitrate:

Test Type	:	Local lymph node assay (LLNA)
Species	:	Mouse
Method	:	OECD Test Guideline 429
Result	:	Does not cause skin sensitization.

Imazethapyr:

Test Type	:	Magnussen-Kligman test
Routes of exposure	:	Skin contact
Species	:	Guinea pig
Method	:	OECD Test Guideline 406
Result	:	Not a skin sensitizer.

Germ cell mutagenicity

Based on available data, the classification criteria are not met.

Product:

Genotoxicity in vitro	:	Test Type: Micronucleus test Method: OECD Test Guideline 487 Result: negative Test Type: Ames test Metabolic activation: with and without metabolic activation Method: OECD Test Guideline 471 Result: negative

Components:

clomazone (ISO):

Genotoxicity in vitro : Test Type: Ames test
Test system: Salmonella typhimurium
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 471
Result: negative
GLP: yes

Test system: Chinese hamster ovary cells
Metabolic activation: with and without metabolic activation
Result: negative

Genotoxicity in vivo : Test Type: Cytogenetic assay
Species: Rat
Method: OECD Test Guideline 473
Result: negative

sodium nitrate:

Genotoxicity in vitro : Test Type: Chromosome aberration test in vitro
Method: OECD Test Guideline 473
Result: negative

Genotoxicity in vivo : Test Type: unscheduled DNA synthesis assay
Species: Mouse
Application Route: Oral
Result: negative

Calcium chloride, dihydrate:

Genotoxicity in vitro : Test Type: reverse mutation assay
Metabolic activation: Metabolic activation
Result: negative

Test Type: Chromosome aberration test in vitro
Result: negative

Germ cell mutagenicity - Assessment : In vitro tests did not show mutagenic effects

Imazethapyr:

Genotoxicity in vitro : Test Type: Ames test
Metabolic activation: with and without metabolic activation
Result: negative

Test Type: Ames test
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 471
Result: negative

Genotoxicity in vivo : Test Type: Micronucleus test

Species: mice
Result: negative

Carcinogenicity

Based on available data, the classification criteria are not met.

Components:

clomazone (ISO):

Species	:	Rat, male and female
Application Route	:	Oral
Exposure time	:	2 Years
Result	:	negative

Species	:	Mouse
Method	:	OECD Test Guideline 453
Result	:	negative

Reproductive toxicity

Based on available data, the classification criteria are not met.

Components:

clomazone (ISO):

Effects on fertility	:	Test Type: Two-generation study Species: Rat, male and female Application Route: Oral Result: negative
Effects on fetal development	:	Test Type: Embryo-fetal development Species: Rat Application Route: Oral Symptoms: Maternal effects. Result: negative Test Type: Embryo-fetal development Species: Rabbit Application Route: Oral Symptoms: Maternal effects. Result: negative

sodium nitrate:

Effects on fertility	:	Test Type: reproductive and developmental toxicity study Species: Rat Application Route: Oral Result: negative Remarks: Based on data from similar materials
Effects on fetal development	:	Test Type: reproductive and developmental toxicity study Species: Rat Application Route: Oral Result: negative

Calcium chloride, dihydrate:

Effects on fetal development : Species: Rabbit
Application Route: Oral
Dose: 1.69, 7.85, 35.6, 169 mg/kg/d
Duration of Single Treatment: 13 d
General Toxicity Maternal: NOAEL: > 169 mg/kg bw/day
Embryo-fetal toxicity.: NOAEL: > 169 mg/kg bw/day
Result: negative

Reproductive toxicity - Assessment : Weight of evidence does not support classification for reproductive toxicity

Imazethapyr:

Effects on fetal development : Species: Rat
General Toxicity Maternal: NOEL: 1.250 mg/kg bw/day
Embryo-fetal toxicity.: NOEL: 1.250 mg/kg bw/day
Method: OECD Test Guideline 414

STOT-single exposure

Based on available data, the classification criteria are not met.

STOT-repeated exposure

Based on available data, the classification criteria are not met.

Components:

Calcium chloride, dihydrate:

Assessment : The substance or mixture is not classified as specific target organ toxicant, repeated exposure.

Repeated dose toxicity

Components:

clomazone (ISO):

Species	:	Rat, male and female
NOEL	:	1000 ppm
Application Route	:	Oral
Exposure time	:	90 d
Symptoms	:	increased liver weight

Species	:	Rat
LOAEL	:	400 mg/kg
Exposure time	:	90 d
Method	:	OECD Test Guideline 408
Symptoms	:	Liver effects

Aspiration toxicity

Based on available data, the classification criteria are not met.

Components:

clomazone (ISO):

The substance does not have properties associated with aspiration hazard potential.

Further information

Product:

Remarks : No data available

Components:

clomazone (ISO):

Remarks : When fed to animals, clomazone caused decreased activity, tearing eyes, bleeding from the nose and incoordination.

SECTION 12. ECOLOGICAL INFORMATION

Ecotoxicity

Product:

Toxicity to fish	:	LC50 (Danio rerio (zebra fish)): 494,82 mg/l Exposure time: 96 h Method: OECD Test Guideline 203
Toxicity to daphnia and other aquatic invertebrates	:	EC50 (Daphnia magna (Water flea)): 1.911 mg/l End point: Immobilization Exposure time: 48 h Method: OECD Test Guideline 202 GLP: yes
Toxicity to algae/aquatic plants	:	EyC50 (Pseudokirchneriella subcapitata (algae)): 762 mg/l Exposure time: 72 h Method: OECD Test Guideline 201 GLP: yes
Toxicity to soil dwelling organisms	:	Method: OECD Test Guideline 216 Remarks: This product does not have any known adverse effect on the soil organisms tested. Method: OECD Test Guideline 217 Remarks: This product does not have any known adverse effect on the soil organisms tested. LC50 (Eisenia andrei (red worm)): > 4.000 mg/kg Exposure time: 7 d Method: OECD Test Guideline 207
Toxicity to terrestrial organisms	:	LD50 (Coturnix japonica (Japanese quail)): > 3.300 mg/kg Method: OECD Test Guideline 223 GLP: yes LD50 (Apis mellifera (bees)): 104.06

Exposure time: 48 h
End point: Acute contact toxicity
Method: OECD Test Guideline 214

LD50 (Apis mellifera (bees)): > 2561.64
Exposure time: 48 h
End point: Acute oral toxicity
Method: OECD Test Guideline 213

Components:

clomazone (ISO):

Toxicity to fish : LC50 (Menidia beryllina (Silverside)): 6,3 mg/l
Exposure time: 96 h

LC50 (Oncorhynchus mykiss (rainbow trout)): > 45 mg/l
Exposure time: 96 h

LC50 (Lepomis macrochirus (Bluegill sunfish)): 34 mg/l
Exposure time: 96 h

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): 40,8 mg/l
Exposure time: 48 h

EC50 (Daphnia): 5,2 mg/l
Exposure time: 48 h

EC50 (Daphnia magna (Water flea)): 12,7 mg/l
Exposure time: 48 h
Test Type: static test

EC50 (Mysidopsis bahia (opossum shrimp)): 9,8 mg/l
Exposure time: 48 h

LC50 (Americamysis bahia (mysid shrimp)): 0,57 mg/l
Exposure time: 96 h
Test Type: flow-through test

Toxicity to algae/aquatic plants : EbC50 (Selenastrum capricornutum (green algae)): 2 mg/l
Exposure time: 72 h

ErC50 (Selenastrum capricornutum (green algae)): 4,1 mg/l
Exposure time: 72 h

ErC50 (Navicula pelliculosa (Freshwater diatom)): 0,136 mg/l
Exposure time: 120 h

EC50 (Lemna gibba (duckweed)): 13,9 mg/l
Exposure time: 7 d

NOEC (Navicula pelliculosa (Freshwater diatom)): 0,05 mg/l
End point: Growth rate
Exposure time: 120 h

NOEC (algae): 0,05 mg/l

Exposure time: 96 h

EC50 (Lemna gibba (duckweed)): 13,9 mg/l
Exposure time: 7 d

EC50 (algae): 0,136 mg/l
Exposure time: 72 h

M-Factor (Acute aquatic toxicity) : 1

Toxicity to fish (Chronic toxicity) : NOEC (Oncorhynchus mykiss (rainbow trout)): 2,3 mg/l
Exposure time: 21 d
Test Type: flow-through test

NOEC (Oncorhynchus mykiss (rainbow trout)): 2,29 mg/l
Exposure time: 57 d

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : NOEC (Daphnia magna (Water flea)): 2,2 mg/l
Exposure time: 21 d

NOEC (Americamysis bahia (mysid shrimp)): 0,032 mg/l
Exposure time: 28 d
Test Type: flow-through test

NOEC (Daphnia magna (Water flea)): 1,25 mg/l
Exposure time: 21 d
Test Type: static test

M-Factor (Chronic aquatic toxicity) : 1

Toxicity to soil dwelling organisms : LC50 (Eisenia fetida (earthworms)): 391,2 mg/kg
Exposure time: 14 d

Toxicity to terrestrial organisms : LD50 (Anas platyrhynchos (Mallard duck)): > 2.510 mg/kg

LC50 (Anas platyrhynchos (Mallard duck)): > 5620 ppm
Remarks: Dietary

LD50 (Coturnix japonica (Japanese quail)): > 2000

NOEC (Colinius virginianus): 94 mg/kg
End point: Reproduction Test

LC50 (Apis mellifera (bees)): > 85.29

LC50 (Apis mellifera (bees)): > 100
Remarks: Contact

sodium nitrate:

Toxicity to fish : LC50 (Oncorhynchus mykiss (rainbow trout)): > 100 mg/l

		Exposure time: 96 h Method: OECD Test Guideline 203 Remarks: Based on data from similar materials
Toxicity to daphnia and other aquatic invertebrates	:	EC50 (Daphnia magna (Water flea)): 8.600 mg/l Exposure time: 24 h Method: OECD Test Guideline 202
Toxicity to fish (Chronic toxicity)	:	NOEC (Pimephales promelas (fathead minnow)): 157 mg/l Exposure time: 32 d
Toxicity to microorganisms	:	EC50: > 1.000 mg/l Exposure time: 3 h Method: OECD Test Guideline 209

Calcium chloride, dihydrate:

Toxicity to fish	:	LC50 (Pimephales promelas (fathead minnow)): 4.630 mg/l Exposure time: 96 h Test Type: static test
Toxicity to daphnia and other aquatic invertebrates	:	LC50 (Daphnia magna (Water flea)): 2.400 mg/l Exposure time: 48 h Method: OECD Test Guideline 202
Toxicity to algae/aquatic plants	:	EC50 (Pseudokirchneriella subcapitata (algae)): 2.900 mg/l Exposure time: 72 h Method: OECD Test Guideline 201
Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity)	:	EC50 (Daphnia magna (Water flea)): 610 mg/l Exposure time: 21 d

Imazethapyr:

Toxicity to fish	:	LC50 (Pimephales promelas (fathead minnow)): 190 mg/l Exposure time: 96 h
Toxicity to daphnia and other aquatic invertebrates	:	EC50 (Daphnia similis (Water flea)): 127 mg/l Exposure time: 48 h NOEC (Ceriodaphnia dubia (water flea)): 80 mg/l Exposure time: 48 h
Toxicity to algae/aquatic plants	:	EC50 (Selenastrum capricornutum (green algae)): 73 mg/l Exposure time: 96 h EC50 (Lemna gibba (duckweed)): 0,101 mg/l Exposure time: 14 h
Toxicity to fish (Chronic toxicity)	:	NOEC (Pimephales promelas (fathead minnow)): 60 mg/l Exposure time: 96 h
Toxicity to soil dwelling organisms	:	LC50 (Eisenia fetida (earthworms)): 8.900 mg/kg Exposure time: 14 d

Toxicity to terrestrial organisms : LD50 (Apis mellifera (bees)): > 20 µg/bee
End point: Acute contact toxicity

LD50 (Coturnix japonica (Japanese quail)): > 2.000 mg/kg

Persistence and degradability

Components:

clomazone (ISO):

Biodegradability : Result: Not readily biodegradable.
Remarks: Substance/product is moderately persistent in the environment.
Primary degradation half-lives vary with circumstances, from a few weeks to a few months in aerobic soil and water.

sodium nitrate:

Biodegradability : Remarks: The methods for determining biodegradability are not applicable to inorganic substances.

Imazethapyr:

Biodegradability : Result: Not readily biodegradable.

Bioaccumulative potential

Product:

Bioaccumulation : Remarks: No data available

Components:

clomazone (ISO):

Bioaccumulation : Bioconcentration factor (BCF): 27 - 40
Remarks: Low potential for bioaccumulation

Partition coefficient: n-octanol/water : log Pow: 2,365 (20 °C)
Method: OECD Test Guideline 107

log Pow: 2,61 - 2,69 (20 - 21 °C)
pH: 4 - 10
Method: Regulation (EC) No. 440/2008, Annex, A.8

Imazethapyr:

Bioaccumulation : Bioconcentration factor (BCF): 0,1
Remarks: Does not bioaccumulate.

Partition coefficient: n-octanol/water : log Pow: 0,98
pH: 5

Mobility in soil

Components:

clomazone (ISO):

Distribution among environmental compartments	:	Koc: 300 ml/g, log Koc: 2,47 Remarks: Moderately mobile in soils
Stability in soil	:	

Imazethapyr:

Distribution among environmental compartments	:	Remarks: Moderately mobile in soils

Other adverse effects

Product:

Additional ecological information	:	An environmental hazard cannot be excluded in the event of unprofessional handling or disposal. Toxic to aquatic life with long lasting effects.

Components:

clomazone (ISO):

Additional ecological information	:	An environmental hazard cannot be excluded in the event of unprofessional handling or disposal. Very toxic to aquatic life with long lasting effects.

SECTION 13. DISPOSAL CONSIDERATIONS

Disposal methods

Waste from residues	:	The product should not be allowed to enter drains, water courses or the soil. Do not contaminate ponds, waterways or ditches with chemical or used container. Send to a licensed waste management company.
Contaminated packaging	:	It is prohibited to reuse, bury, burn or sell packaging. Washable packaging: Triple wash packs of less than 20 liters and pressure wash packs of 20 liters or more. Triple Wash (Manual Wash): Completely empty the contents of the package into the sprayer tank, keeping it in an upright position for 30 seconds; Add clean water to the package up to ¼ of its volume; Cover the package well and shake it for 30 seconds; Pour the wash water into the spray tank; Do this operation three times; Make the plastic or metal packaging unusable by perforating the bottom. Pressure wash: Fit the empty package in the appropriate place of the funnel installed on the sprayer; Activate the mechanism to release the water jet; Direct the water jet to all

the inside walls of the package, for 30 seconds; Wash water must be transferred to the sprayer tank; Make the plastic or metal packaging unusable by perforating the bottom. In both procedures, puncture the container at its base without damaging the label. Within a period of up to one year from the date of purchase, the user must return the empty packaging, with lid, to the establishment where the product was purchased or to the place indicated on the invoice, issued at the time of purchase.Activate the mechanism to release the water jet. Direct the water jet to all the inside walls of the package, for 30 seconds. Wash water must be transferred to the sprayer tank. Make the plastic or metal packaging unusable by perforating the bottom.

SECTION 14. TRANSPORT INFORMATION

International Regulations

UNRTDG

UN number	:	UN 3082
Proper shipping name	:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (Clomazone)
Class	:	9
Packing group	:	III
Labels	:	9
Environmentally hazardous	:	yes

IATA-DGR

UN/ID No.	:	UN 3082
Proper shipping name	:	Environmentally hazardous substance, liquid, n.o.s. (Clomazone)
Class	:	9
Packing group	:	III
Labels	:	Miscellaneous
Packing instruction (cargo aircraft)	:	964
Packing instruction (passenger aircraft)	:	964
Environmentally hazardous	:	yes

IMDG-Code

UN number	:	UN 3082
Proper shipping name	:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (Clomazone)
Class	:	9
Packing group	:	III
Labels	:	9
EmS Code	:	F-A, S-F
Marine pollutant	:	yes

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not applicable for product as supplied.

Domestic regulation

ANTT

UN number : UN 3082
Proper shipping name : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (Clomazone)

Class : 9
Packing group : III
Labels : 9
Hazard Identification Number : 90

Special precautions for user

The transport classification(s) provided herein are for informational purposes only, and solely based upon the properties of the unpackaged material as it is described within this Safety Data Sheet. Transportation classifications may vary by mode of transportation, package sizes, and variations in regional or country regulations.

SECTION 15. REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

Law No. 14,785 of December 27, 2023. Decree 4,074 of January 4, 2002 and its regulatory standards. ANTT Resolution No. 5,998/22 of November 3, 2022. This SDS was prepared in accordance with the criteria of ABNT NBR 14725. The user is recommended to pay attention to local regulations.
National List of Carcinogenic Agents for Humans - (LINACH)

Group 2A: Probably carcinogenic to humans
sodium nitrate 7631-99-4

Brazil. List of chemicals controlled by the Federal Police : Not applicable

The ingredients of this product are reported in the following inventories:

TCSI : Not in compliance with the inventory

TSCA : Product contains substance(s) not listed on TSCA inventory.

AIIC : Not in compliance with the inventory

DSL : This product contains the following components that are not on the Canadian DSL nor NDSL.

clomazone (ISO)
Imazethapyr
o-Chlorobenzaldehyde

ENCS : Not in compliance with the inventory

ISHL : Not in compliance with the inventory

KECI : Not in compliance with the inventory

PICCS : Not in compliance with the inventory

IECSC : Not in compliance with the inventory

NZIoC : Not in compliance with the inventory

TECI : Not in compliance with the inventory

SECTION 16. OTHER INFORMATION

Revision Date : 19.02.2026

Date format : dd.mm.yyyy

Full text of other abbreviations

AIIC - Australian Inventory of Industrial Chemicals; ANTT - National Agency for Transport by Land of Brazil; ASTM - American Society for the Testing of Materials; bw - Body weight; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; ERG - Emergency Response Guide; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; Nch - Chilean Norm; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NOM - Official Mexican Norm; NTP - National Toxicology Program; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; TCSI - Taiwan Chemical Substance Inventory; TDG - Transportation of Dangerous Goods; TECI - Thailand Existing Chemicals Inventory; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative; WHMIS - Workplace Hazardous Materials Information System

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